Web Programming Laboratory

Experiment 7 - Recitation

Question 1

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>React Portfolio Components</title> <script
src="https://unpkg.com/react@18/umd/react.development.js"></script> <script
src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script> <script
src="https://unpkg.com/@babel/standalone/babel.min.js"></script> <style> * { margin: 0; padding: 0;
box-sizing: border-box; } body { font-family: 'Segoe UI', sans-serif; background: #f5f7fa; } .header {
background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%); color: white; padding: 60px 20px; text-align:
center; box-shadow: 0 5px 20px rgba(0,0,0,0.15); } .header h1 { font-size: 3em; margin-bottom: 10px; }
.header p { font-size: 1.3em; opacity: 0.95; } .section { max-width: 1000px; margin: 40px auto; padding:
35px; background: white; border-radius: 15px; box-shadow: 0 5px 25px rgba(0,0,0,0.08); } .section h2 { color:
#f5576c; border-bottom: 3px solid #f093fb; padding-bottom: 12px; margin-bottom: 25px; font-size: 2em; }
.section p { line-height: 1.8; color: #555; font-size: 1.1em; } .skills-grid { display: flex; gap: 15px;
flex-wrap: wrap; margin-top: 10px; } .skill-badge { background: linear-gradient(135deg, #f093fb 0%, #f5576c
100%); color: white; padding: 12px 24px; border-radius: 25px; font-size: 1em; font-weight: 600; box-shadow: 0
3px 10px rgba(245, 87, 108, 0.3); transition: transform 0.3s ease; } .skill-badge:hover { transform:
translateY(-3px); box-shadow: 0 5px 15px rgba(245, 87, 108, 0.4); } .project-card { border: 2px solid
#f0f2f5; border-radius: 12px; padding: 25px; margin: 15px 0; transition: all 0.3s ease; } .project-card:hover
{ border-color: #f093fb; transform: translateY(-5px); box-shadow: 0 8px 25px rgba(240, 147, 251, 0.2); }
.project-card h3 { color: #2c3e50; font-size: 1.5em; margin-bottom: 10px; } .project-card p { color: #666;
margin: 10px 0; line-height: 1.6; } .project-card small { color: #f5576c; font-weight: 600; background:
#ffe5ec; padding: 5px 12px; border-radius: 8px; } .footer { text-align: center; padding: 30px; background:
#2c3e50; color: white; margin-top: 40px; font-size: 1.1em; } </style> </head> <body> <div id="root"></div>
<script type="text/babel"> // Functional Component: Header function Header({ name, title }) { return (
<header className="header"> <h1>■■■ {name}</h1> <p>{title}</p> </header> ); } // Functional Component: About
function About() { return ( <section className="section"> <h2>■ About Me</h2> <p>I am a second-year AIML
engineering student at SFIT, passionate about machine learning, web development, and building intelligent
systems. My journey combines academic excellence with hands-on projects that solve real-world problems.</p>
</section> ); }
```

```
// Functional Component: Skills function Skills({ skills }) { return ( <section className="section"> <h2>■
Skills</h2> <div className="skills-grid"> {skills.map((skill, i) => <span key={i}
className="skill-badge">{skill}</span>)} </div> </section> ); } // Class Component: Projects class Projects
extends React.Component { render() { return ( <section className="section"> <h2>■ Projects</h2>
{this.props.projects.map((proj, i) => ( <div key={i} className="project-card"> <h3>{proj.name}</h3>
<p>{proj.description}</p> <small>■ Tech: {proj.tech}</small> </div> ))} </section> ); } } // Functional
Component: Footer function Footer() { return <footer className="footer"><p>© 2026 Tony Rodrigues | Built with
React ■■</p></footer>; } // Main App function App() { const skills = ['Python', 'Machine Learning', 'React',
'HTML/CSS', 'JavaScript', 'SQL', 'NumPy', 'Pandas', 'TensorFlow', 'Git']; const projects = [ { name: 'Disease
Predictor', description: 'ML-based disease prediction system using Naive Bayes and Neural Networks with 92%
accuracy.', tech: 'Python, Scikit-learn, Flask' }, { name: 'Portfolio Website', description: 'Responsive
personal portfolio built with React components and modern design principles.', tech: 'React, HTML, CSS,
JavaScript' }, { name: 'Data Pipeline', description: 'End-to-end ML pipeline with data preprocessing, feature
engineering, and model evaluation.', tech: 'Python, Pandas, Sklearn, Matplotlib' } ]; return ( <div> <Header
name="Tony Rodrigues" title="SE AIML Student | Web Developer | ML Enthusiast" /> <About /> <Skills
skills={skills} /> <Projects projects={projects} /> <Footer /> </div> ); }
```

```
ReactDOM.createRoot(document.getElementById('root')).render(<App />); </script> </body> </html>
```

Output

👨‍💻 Tony Rodrigues

SE AIML Student | Web Developer | ML Enthusiast

📖 About Me

I am a second-year AIML engineering student at SFIT, passionate about machine learning, web development, and building intelligent systems. My journey combines academic excellence with hands-on projects that solve real-world problems.

🚀 Skills

Python | Machine Learning | React | HTML/CSS | JavaScript | SQL | NumPy | Pandas | TensorFlow | Git

💼 Projects

Disease Predictor

ML-based disease prediction system using Naive Bayes and Neural Networks with 92% accuracy.

💻 Tech: Python, Scikit-learn, Flask

Portfolio Website

Responsive personal portfolio built with React components and modern design principles.

💻 Tech: React, HTML, CSS, JavaScript

Data Pipeline

End-to-end ML pipeline with data preprocessing, feature engineering, and model evaluation.

💻 Tech: Python, Pandas, Sklearn, Matplotlib

© 2026 Tony Rodrigues | Built with React ⚛️

Question 2

Source Code

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0"> <title>React Props - Dynamic Portfolio</title> <script
src="https://unpkg.com/react@18/umd/react.development.js"></script> <script
src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script> <script
src="https://unpkg.com/@babel/standalone/babel.min.js"></script> <style> * { margin: 0; padding: 0;
box-sizing: border-box; } body { font-family: 'Segoe UI', sans-serif; background: linear-gradient(135deg,
#ffecd2 0%, #fcb69f 100%); min-height: 100vh; } .header { background: linear-gradient(135deg, #FF6B6B 0%,
#FFE66D 100%); color: white; padding: 60px 20px; text-align: center; box-shadow: 0 5px 20px rgba(0,0,0,0.15);
} .header h1 { font-size: 3em; margin-bottom: 10px; } .header p { font-size: 1.3em; opacity: 0.95; } .section
{ max-width: 1000px; margin: 40px auto; padding: 35px; background: white; border-radius: 15px; box-shadow: 0
5px 25px rgba(0,0,0,0.08); } .section h2 { color: #FF6B6B; border-bottom: 3px solid #FFE66D; padding-bottom:
12px; margin-bottom: 25px; font-size: 2em; } .section p { line-height: 1.8; color: #555; font-size: 1.1em; }
.profile-card { display: flex; gap: 25px; align-items: center; padding: 25px; background:
linear-gradient(135deg, #f8f9fa 0%, #fff 100%); border-radius: 15px; margin: 15px 0; transition: all 0.3s
ease; border: 2px solid #f0f2f5; } .profile-card:hover { transform: translateY(-5px); box-shadow: 0 10px 30px
rgba(255, 107, 107, 0.15); border-color: #FFE66D; } .profile-card .avatar { width: 90px; height: 90px;
border-radius: 50%; background: linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%); display: flex;
align-items: center; justify-content: center; color: white; font-size: 2.2em; font-weight: bold; box-shadow:
0 5px 15px rgba(255, 107, 107, 0.3); } .profile-card .info h3 { color: #2c3e50; margin-bottom: 8px;
font-size: 1.4em; } .profile-card .info p { color: #666; font-size: 1.05em; } .footer { text-align: center;
padding: 30px; background: #2c3e50; color: white; margin-top: 40px; font-size: 1.1em; } </style> </head>
<body> <div id="root"></div> <script type="text/babel"> // Functional Component with Props: Profile function
Profile({ name, role }) { return ( <header className="header"> <h1>■ {name}</h1> <p>{role}</p> </header> ); }
// Class Component with Props: About class About extends React.Component { render() { return ( <section
className="section"> <h2>■ About Me</h2> <p>{this.props.description}</p> </section> ); } } // Functional
Component with Props: TeamMember
```

```
function TeamMember({ name, role, initial }) { return ( <div className="profile-card"> <div
className="avatar">{initial}</div> <div className="info"> <h3>{name}</h3> <p>■ {role}</p> </div> </div> ); }
// Team Section using Props function Team({ members }) { return ( <section className="section"> <h2>■
Team</h2> {members.map((m, i) => ( <TeamMember key={i} name={m.name} role={m.role} initial={m.name[0]} />
))} </section> ); } function Footer({ year }) { return <footer className="footer"><p>© {year} Tony Rodrigues |
Props Demo with React ■■</p></footer>; } // App: Passing Props from Parent to Children function App() { const
teamMembers = [ { name: "Tony Rodrigues", role: "Lead Developer & ML Engineer" }, { name: "Amit Kumar", role:
"UI/UX Designer" }, { name: "Sarah Chen", role: "Backend Developer" }, { name: "Mike Johnson", role: "Data
Scientist" } ]; return ( <div> <Profile name="Tony Rodrigues" role="SE AIML Student | Web Developer | React
Enthusiast" /> <About description="I am a second-year AIML engineering student at SFIT, passionate about
machine learning and web development. This page demonstrates passing data between React components using
props - a core concept in React that enables component communication and reusability." /> <Team
members={teamMembers} /> <Footer year={2026} /> </div> ); }
ReactDOM.createRoot(document.getElementById('root')).render(<App />); </script> </body> </html>
```

Output

👤 Tony Rodrigues

SE AIML Student I Web Developer I React Enthusiast

📖 About Me

I am a second-year AIML engineering student at SFIT, passionate about machine learning and web development. This page demonstrates passing data between React components using props - a core concept in React that enables component communication and reusability.

👥 Team

T — **Tony Rodrigues**
🎯 Lead Developer & ML Engineer

A — **Amit Kumar**
🎯 UI/UX Designer

S — **Sarah Chen**
🎯 Backend Developer

M — **Mike Johnson**
🎯 Data Scientist

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